

2110531 Data Science and Data Engineering Tools (2026/1)

Take-Home Midterm Exam Report

Bangkok CCTV Vehicle Detection

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Source code, CSVs, experiment outputs (GitHub): https://github.com/RiewS/DSDE_Midterm_-Bangkok-CCTV-Vehicle-Detection

Model weights and prepared data (Google Drive): <Google Drive link>

Chapter 1: Introduction

The task is multi-class object detection of vehicles in Bangkok traffic-camera (CCTV) images. The images were scraped automatically every 30 minutes from the Bangkok Metropolitan Administration (BMA) traffic platform (cpudapp.bangkok.go.th/bmatraffic) between 25/08/2026 06:00 and 31/08/2026 21:00. After a quality screening, 20 cameras were kept. Every vehicle has to be localised with a bounding box and assigned to one of eight classes: Car, Motorcycle, Bus, Truck, Tuktuk, Van, Pickup and Songthaew.

Submissions are scored on Kaggle with **mAP@50**: the COCO-style average precision at IoU 0.50, computed with pycocotools for each class and averaged over the eight classes. Because every class has the same weight, rare classes such as Songthaew or Tuktuk affect the score as much as Car does.

Objectives of this work:

- explore and clean the data, and build a validation scheme that reflects the hidden test set;
- fine-tune a modern pretrained detector (YOLO26 from the Ultralytics framework) and evaluate it with the same metric as Kaggle;
- analyse the errors and describe how the model could be improved.

The code is in two Jupyter notebooks: `vehicle_detection.ipynb` (EDA, cleaning, split, YOLO26s experiments, evaluation, visualisation, submission; local GPU) and `cv_kfold_kaggle.ipynb` / `cv_kfold_kaggle_v2.ipynb` (final YOLO26m camera 5-fold cross-validation v1 / v2; Kaggle GPU), plus `experiments/ensemble_wbf.py` for the final 10-model ensemble. No generative-AI / VLM API was used, so there is no API cost. The final submission (an ensemble of 10 YOLO26m models) reaches **0.630 mAP@50** on the public leaderboard.

Chapter 2: Data Preparation

The analysis follows the data-preparation steps taught in the course (Week 2 slides and Lab1-Lab4): (1) examine the data set, (2) understand / narrow down the features and check for leakage, (3) handle outliers and skewness, and (4) build the train / validation split with a fixed random seed.

2.1 Examining the data set

Table 1: Dataset summary

Item	Value
Train images	2,991 (15 cameras)
Test images	1,013 (5 cameras); 997 are listed in <code>sample_submission.csv</code>
Image size	352 x 288 pixels (all images)
Annotated boxes (train)	27,396 in 2,889 images; 102 images have no vehicle
Boxes per image	mean 9.2, max 48
Box size (w x h)	median 13 x 20 px, 25% of boxes are smaller than 8 x 12 px
Submission format	id, image_id, class_id, confidence, x1, y1, x2, y2 (one row per box)

Table 2: Data dictionary of `train.csv` (27,396 rows x 7 columns)

Column	Type	Description	Missing
id	int	row id - identifier only, not a feature	0

image_id	str	<camera>_<YYYYMMDD>_<HHMMSS>.jpg; camera, date, hour and day/night are derived from it	0
class_id	int (nominal)	target class 0-7	0
x1, y1	int	top-left corner of the box [px]	0
x2, y2	int	bottom-right corner of the box [px]	0

Out-of-range values and miscodes. No missing values; all class ids are in 0-7; no inverted boxes and no box outside the image; every image_id exists on disk and all images are 352 x 288. Found: 25 exact duplicate boxes, 15 boxes with a side < 2 px, and 102 training images without any vehicle (kept as negative examples).

Target distribution

Table 3: Class distribution in the training set

ID	Class	Boxes	Share	Images containing class	Imbalance vs Car
0	Car	17,762	64.83%	2,712	1.0x
1	Motorcycle	6,563	23.96%	2,160	2.7x
2	Bus	482	1.76%	410	36.9x
3	Truck	1,356	4.95%	962	13.1x
4	Tuktuk	265	0.97%	226	67.0x
5	Van	354	1.29%	294	50.2x
6	Pickup	497	1.81%	408	35.7x
7	Songthaew	117	0.43%	59	151.8x

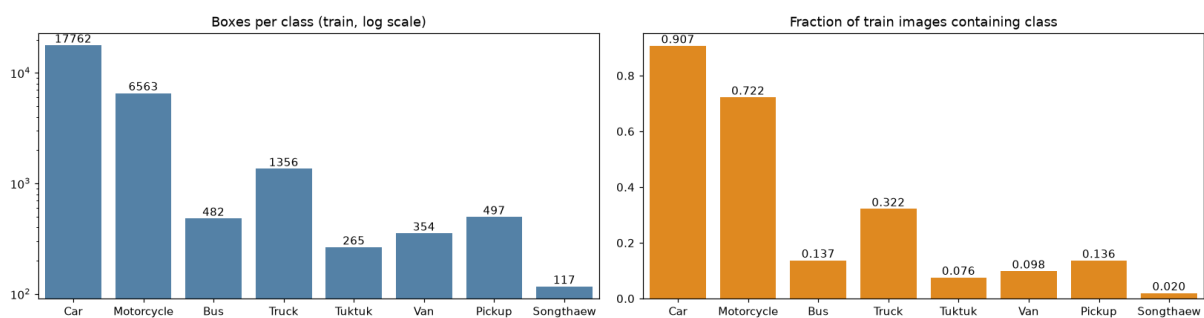


Figure 1: Boxes per class (log scale) and fraction of training images that contain each class

The target is heavily imbalanced: Car is 64.8% of all boxes while Songthaew is 0.43% (152 times fewer). Because mAP@50 averages the AP of the eight classes equally, the rare classes matter as much as Car.

2.2 Feature understanding, association with the target and leakage

Camera. Train uses cameras 1066, 12, 1407, 1426, 1427, 1437, 172, 180, 182, 214, 222, 229, 231, 232, 244; test uses 1068, 1072, 1192, 1439, 227. **The two sets are disjoint**, so the model must generalise to unseen viewpoints.

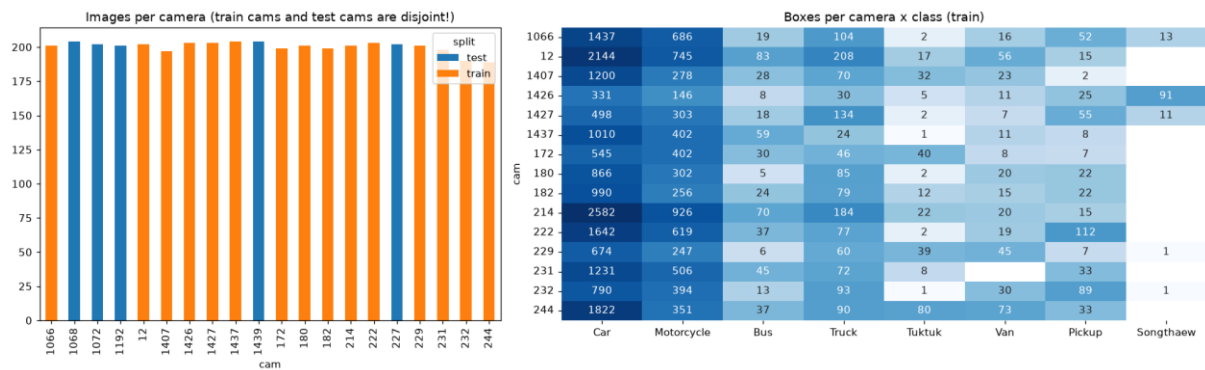


Figure 2: Images per camera (train vs test) and boxes per camera x class

Chi-square test of camera x class (slide appendix): $\chi^2 = 4,823$, $\text{dof} = 98$, $p < 1e-300$, Cramer's $V = 0.159$. H_0 (same class distribution for every camera) is rejected: the vehicle mix depends strongly on the camera. The standardised residuals show, for example, that 91 of the 117 Songthaews come from camera 1426 (residual +53), Pickups are over-represented at cameras 232 and 222 and Tuktuks at 244, 229 and 172.

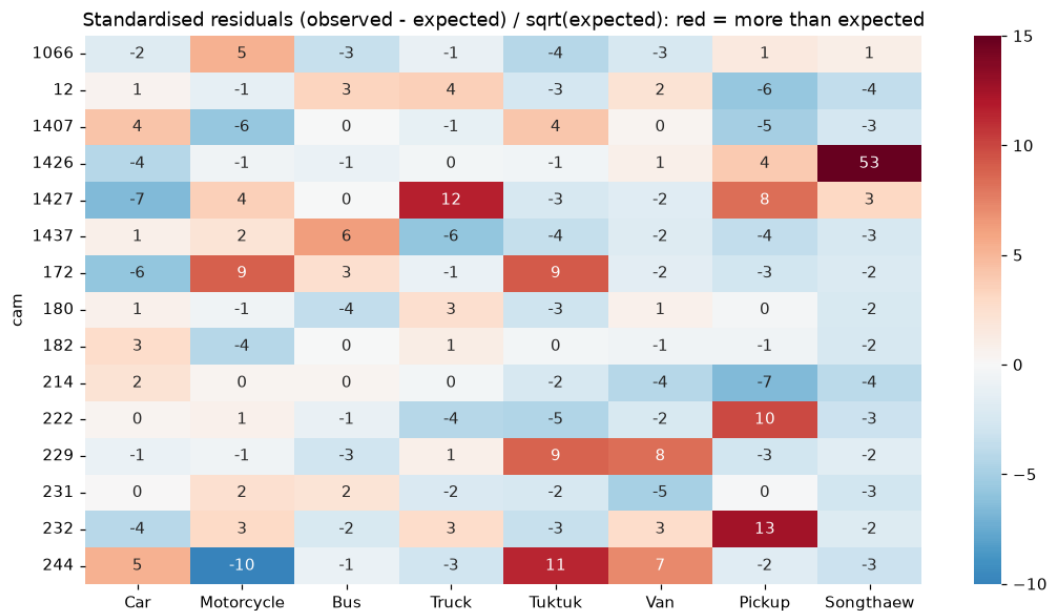


Figure 3: Standardised chi-square residuals, camera x class (red = more boxes than expected)

Time of day. Frames cover 06:00-21:00 every 30 minutes. Street lights keep night frames bright, so night was defined by clock time (after 18:30, about 17% of frames). The class mix also changes at night (χ^2 -square $p = 6.1e-33$, Cramer's $V = 0.078$); e.g. Songthaew rises from 0.27% of boxes by day to 1.53% at night and Tuktuk from 0.84% to 1.86%.

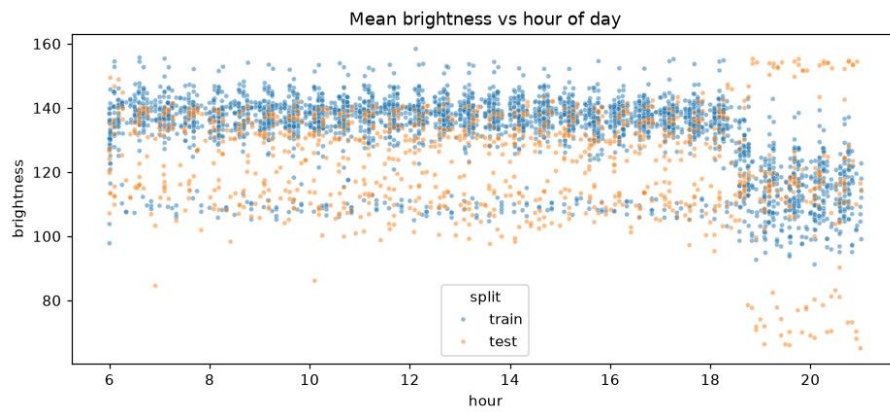


Figure 4: Mean brightness vs hour of day

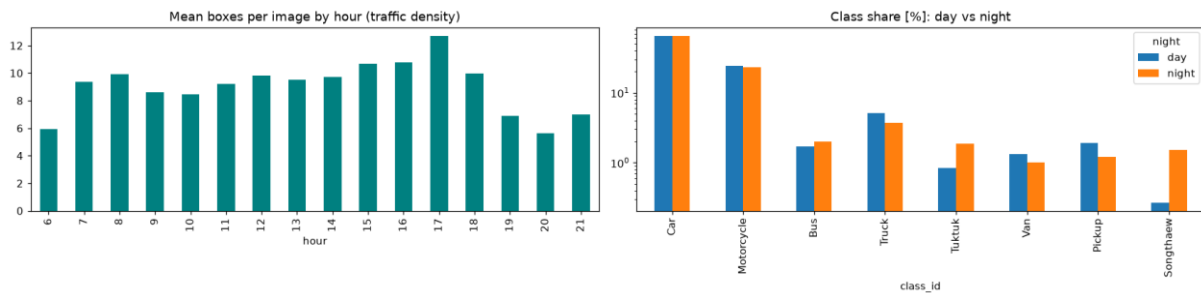


Figure 5: Traffic density by hour and class share by day / night

Leakage check. In Lab1, columns that leak information were removed. Here the leak is at the image level: consecutive frames of the same camera differ by a median of 14.5 grey levels, versus 53.6 for frames of different cameras (3.7x more similar). A random train / validation split would put almost identical scenes on both sides and measure memorisation instead of generalisation. Therefore **whole cameras are held out** for validation (Section 2.6).

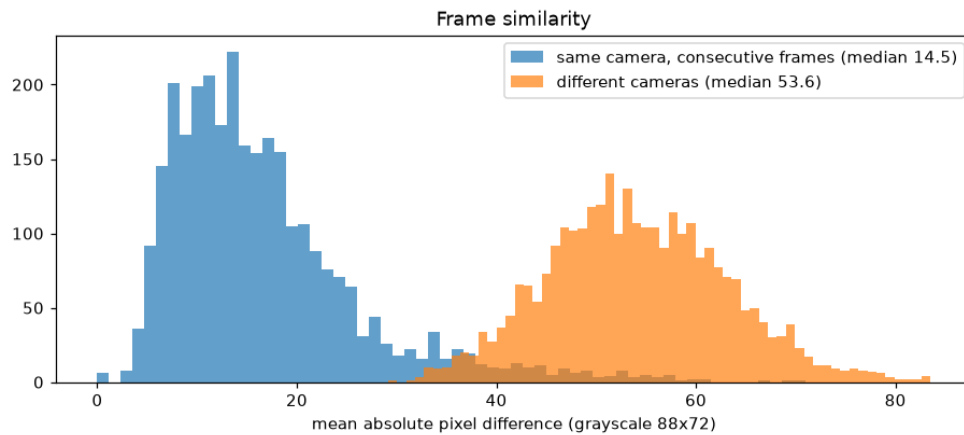


Figure 6: Frame similarity: same camera (consecutive frames) vs different cameras

Train vs test distribution. The test cameras have a similar share of night frames (17.6% vs 17.1%) and a slightly darker mean brightness (123 vs 132).

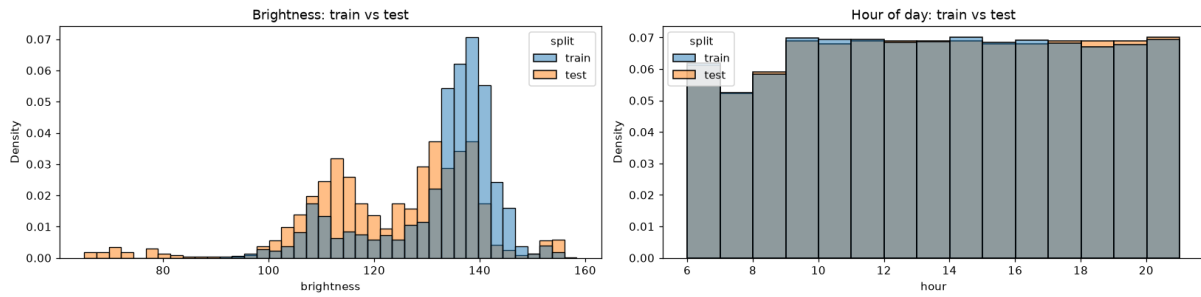


Figure 7: Brightness and hour-of-day distributions of train and test images

2.3 Outliers and skewness

Box sizes are tiny and strongly right-skewed: the box area has skew 7.05 and kurtosis 96.0; after a log transform the skew is 0.19 (Lab4 approach). This motivates upscaling the images (imgsz 640 -> 960) and a multi-scale detector.

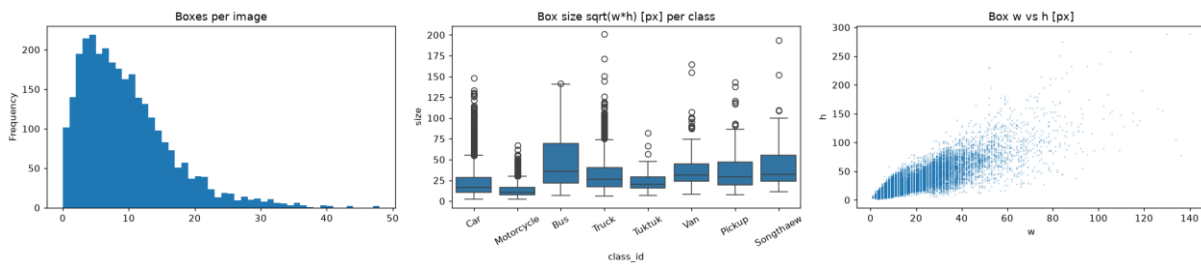


Figure 8: Boxes per image, box size per class and box width vs height

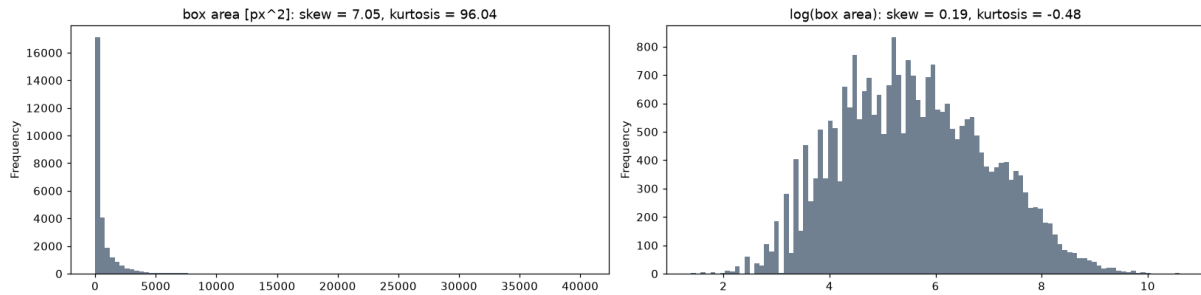


Figure 9: Box area before and after log transform

Table 4: Outliers per class (1.5 x IQR rule)

Class	Boxes	Median area	log-area IQR outliers	Median h/w	h/w IQR outliers
Car	17762	276	5	1.33	452
Motorcycle	6563	119	14	2.56	58
Bus	482	1272	0	1.8	6
Truck	1356	670	4	1.78	13
Tuktuk	265	420	1	1.93	0
Van	354	990	4	1.6	2
Pickup	497	840	0	1.42	1
Songthaew	117	1054	0	1.54	14

The extremes were inspected visually before deciding what to do. The largest boxes are real vehicles very close to camera 1426 (6 boxes are almost full height) and are kept. The extreme aspect ratios are

annotation errors: slivers such as 1×8 or 2×16 px with no vehicle inside (195 boxes have a side ≤ 2 px). Boxes with a side < 2 px are removed; truncating coordinates, as done for numeric features in Lab4, is not meaningful for boxes.



Figure 10: Most extreme boxes: largest area (top), tallest (middle), widest (bottom)

2.4 Visual inspection



Figure 11: Preview: one daytime frame per training camera with ground-truth boxes



Figure 12: Preview: night frames of training cameras. Training uses all frames (day and night); only this preview is split by time

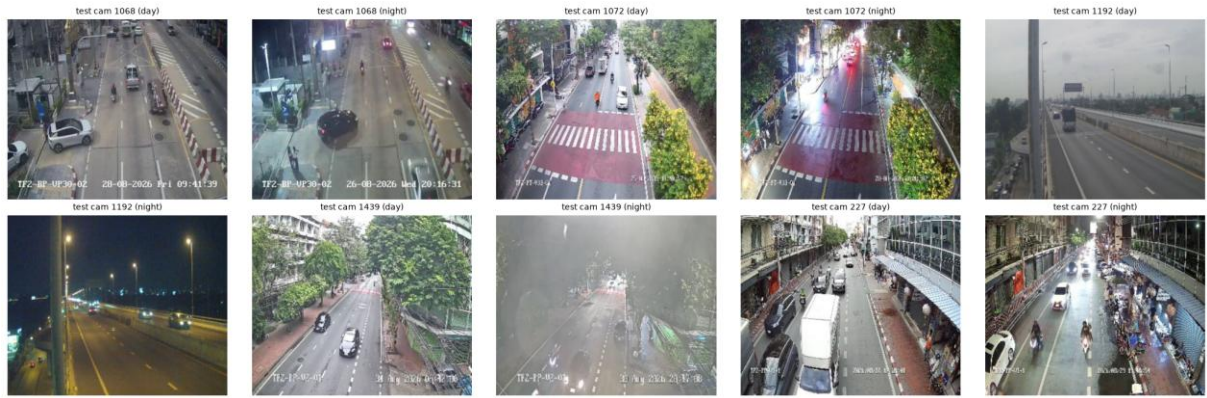


Figure 13: Test cameras (day and night); these viewpoints never appear in training

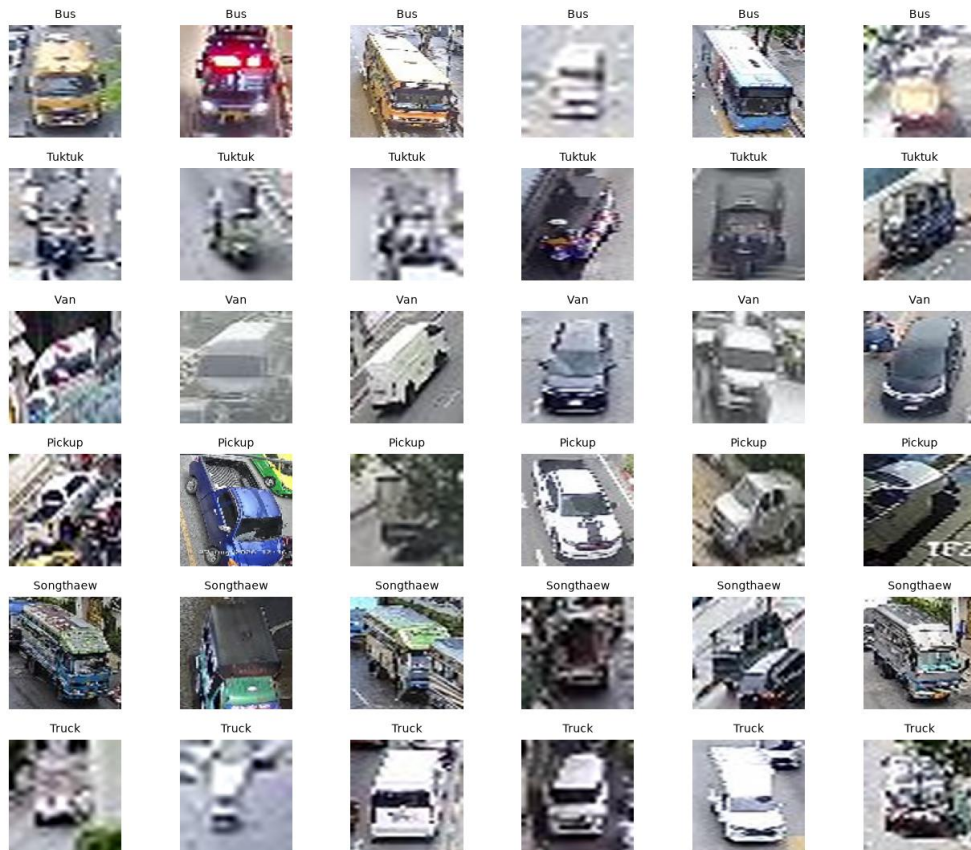


Figure 14: Examples of rare classes (crops). Van / Pickup / Truck are hard to separate even for a human

sample_submission.csv uses different ids from the test file names. It lists 997 long ids (dataset_<cam>_annotated_coco1.0_<cam>_<Thai location name>_<date>_<time>.jpg), while the 1,013 test images are named <cam>_<date>_<time>.jpg. The Kaggle solution uses the file names (Section 4.3), so predictions are written for all 1,013 test images with the file name as image_id.

2.5 Data cleaning

Cleaning is applied to the **training labels only**. Validation is always scored against the raw labels, the same way Kaggle scores the raw test labels.

Table 5: Data-cleaning log

Step	Count
raw boxes	27396

boxes on missing images	0
boxes outside image (clipped)	0
exact duplicates removed	25
tiny boxes (< 2 px) removed	15
near-duplicates removed (same class, IoU > 0.9)	9
class conflicts kept (different class, IoU > 0.9)	280
clean boxes	27347
background images (no boxes)	102

280 pairs of boxes overlap with IoU > 0.9 but have different classes (mostly Car vs Truck and Bus vs Truck): the same vehicle was labelled twice. They were kept because the test labels probably contain the same pattern, and a model that outputs both classes at lower confidence can still match both.

Label audit: rules (Lab5) and model-assisted check

The rule-based cleaning above only removes boxes that are certainly broken. To check the remaining labels, two audits were run (experiments/label_audit.py) and the flagged cases were inspected visually; nothing was changed in the training data afterwards, so the submitted models stay reproducible.

(a) Outlier rules from Lab5 (z-score > 3 and 1.5 x IQR on log aspect ratio and log area, per class):

Table 6: Boxes flagged by the Lab5 outlier rules

Class	z>3 aspect	z>3 area	IQR aspect	IQR area	side <= 2 px
Car	181	3	368	5	6
Motorcycle	31	5	56	14	189
Bus	6	0	13	0	0
Truck	11	2	22	4	0
Tuktuk	0	1	0	1	0
Van	0	2	6	4	0
Pickup	0	0	0	0	0
Songthaew	3	0	12	0	0



Figure 15: Boxes with $|z(\log \text{ aspect})| > 3$: mostly vehicles cut by the image border (valid), a few slivers (errors)

Visual inspection shows that most rule outliers are **valid**: vehicles cut by the image border (only the roof is visible) and thin, head-on motorcycles (163 of the 172 boxes with a side ≤ 3 px and ratio ≥ 4 are Motorcycles). Only a handful are clear annotation errors (e.g. 2×16 px or 7×1 px "Car" slivers). Removing every rule outlier, as for the numeric features of Lab5, would therefore delete hundreds of real vehicles, which also appear in the test set.

(b) Model-assisted audit. The out-of-fold predictions of the 5-fold CV are made by models that never saw the camera, so strong disagreements between model and label point to label problems:

Table 7: Model-assisted label audit (OOF predictions of YOLO26m)

Check	Rule	Flagged	Visual review of the top/random 24
Possible missing label	prediction conf ≥ 0.7 , IoU < 0.3 with every label	674	about 20 of 24 are real unlabelled vehicles, mostly cut by the image border or very close to the camera
Possible wrong class	best prediction (IoU ≥ 0.7) has another class, conf ≥ 0.6	1265	about 3 of 4 look like label errors: vans labelled Truck / Bus, pickups labelled Car
Possible bad box	no prediction with IoU ≥ 0.3	1744	mostly correct labels of tiny, distant vehicles in dense traffic (median 10×14 px): a model weakness, not a label error

Most frequent class disagreements (label $\rightarrow$ model): Truck $\rightarrow$ Car (269), Pickup $\rightarrow$ Car (202), Car $\rightarrow$ Pickup (103), Van $\rightarrow$ Car (87), Car $\rightarrow$ Truck (82), Car $\rightarrow$ Van (67). Part of the Truck $\rightarrow$ Car count comes from the 212 Car/Truck double labels found in Section 2.5. Possible missing labels touch the image border in 23% of the cases, versus 9% of all labels: vehicles that are only partly visible were labelled inconsistently.



Figure 16: Possible missing labels: confident predictions (red) where no label exists (green = existing labels)



Figure 17: Possible wrong classes: label (green) vs. confident model class



Figure 18: Labels without any prediction (random sample): mostly tiny distant vehicles

Decision. The labels contain noise of the same kind that the test labels probably contain (missing border vehicles, Van / Pickup / Truck ambiguity), so the training labels were not edited for the submitted models. The audit explains part of the remaining error (Chapter 5) and gives a concrete list for relabelling in future work (experiments/audit_*.csv).

2.6 Train / validation split and dataset construction

To mimic the unseen-camera test set and avoid the leakage shown in 2.2, **4 whole cameras (1066, 1427, 1437, 244) were held out for validation**. As a stratification over cameras, the notebook tries all 1,365 combinations of 4 cameras out of 15 and keeps the one where every class has at least 10 validation boxes and the validation share of each class is closest to 4/15. The random seed is fixed (42) everywhere.

Table 8: Camera-held-out split after cleaning: 2,194 train / 797 val images (class proportions compared as in Lab3)

Class	Train boxes	Val boxes	Val share	Train %	Val %
Car	12969	4765	26.9%	64.87%	64.79%
Motorcycle	4808	1741	26.6%	24.05%	23.67%
Bus	346	133	27.8%	1.73%	1.81%
Truck	1002	352	26.0%	5.01%	4.79%
Tuktuk	180	85	32.1%	0.90%	1.16%
Van	247	107	30.2%	1.24%	1.45%
Pickup	347	148	29.9%	1.74%	2.01%
Songthaew	93	24	20.5%	0.47%	0.33%

Repeat-factor sampling (RFS). mAP weights all classes equally, so images that contain rare classes are repeated during training (LVIS-style): for class c with image frequency f_c , $r_c = \max(1, \sqrt{t / f_c})$ with $t = 0.15$, and each image is repeated $r = \max_c r_c$ times with stochastic rounding. Songthaew images are repeated about 2.8x, Tuk Tuk 1.4x and Van 1.2x. Repeats are only added to the training list, never to validation.

Labels were converted to the YOLO format (class x_center y_center width height, normalised) and the image lists were written as train.txt / val.txt / full.txt, which Ultralytics reads directly.

2.7 Summary: findings and decisions

Table 9: EDA findings and the resulting decisions

Finding	Decision
Test cameras never appear in training; same-camera frames are near-duplicates	Hold out whole cameras for validation
Class mix depends on camera and time (chi-square $p \sim 0$)	Every class must be present in val; RFS for rare classes
Car : Songthaew = 152 : 1	Repeat-factor sampling; per-class AP analysis
Tiny, right-skewed box sizes	Upscale to imsz 960; multi-scale detector
Sliver boxes are label noise; duplicates	Remove side < 2 px and exact / near duplicates
Lab5 outliers are mostly valid (border-cut vehicles, thin motorcycles)	Do not drop rule outliers; inspect visually
OOF audit: missing border vehicles, Van/Pickup/Truck class noise	Keep labels (test has the same noise); list for relabelling
17% night frames in both train and test	Report day / night mAP; HSV augmentation

Chapter 3: Model

3.1 Choice of model

The course slides (Week 5, Object Detection) describe the change from two-stage CNNs (Faster R-CNN) to one-stage YOLO detectors, transformer detectors (DETR / RT-DETR) and, most recently, end-to-end NMS-free YOLO26. The course example notebook fine-tunes YOLOv8 with Ultralytics. This work uses the newest model in that family, **YOLO26**, fine-tuned from COCO-pretrained weights. Two sizes were used: **YOLO26s** for the experiments on the local 4 GB GPU, and **YOLO26m** for the final model, trained on Kaggle GPUs with camera 5-fold cross-validation, twice: v1 (standard augmentation) and v2 (+ cross-camera copy-paste). The final prediction fuses the 10 fold models (v1 + v2).

Table 10: YOLO26 models used

Property	YOLO26s (experiments)	YOLO26m (final)
Framework	Ultralytics 8.4.163, PyTorch 2.8.0	Ultralytics 8.4.163, PyTorch (Kaggle image)
Parameters	9.95 M	21.8 M
FLOPs	22.8 G at 640	75.0 G at 640 / 169.5 G at 960
Pretraining	COCO 2017 detection (80 classes, ~118k images)	same

Backbone	CSP-style CNN with C3k2 blocks, SPPF and C2PSA (attention) block	same design, wider / deeper
Neck	PAN/FPN feature pyramid, strides 8 / 16 / 32 (small vehicles)	same
Head	Anchor-free, end-to-end (one-to-one) head: no NMS, no DFL	same
Transfer	head rows of car, motorcycle, bus, truck initialised from the COCO classes	same

YOLO is a one-stage detector: a single CNN pass predicts, for every location of three feature maps, a box and class scores. YOLO26 trains with a one-to-one assignment so each object gets a single prediction, which removes the non-maximum-suppression step, and it drops the Distribution Focal Loss. According to the Ultralytics documentation it also uses progressive loss balancing (ProgLoss) and small-target-aware label assignment (STAL), which suits this dataset because most vehicles are tiny.

No generative-AI / VLM API was used, so there is no API cost. Compute: the YOLO26s runs used a local laptop GPU; the YOLO26m cross-validations used the free Kaggle GPU quota (2 x NVIDIA T4): v1 8.0 h and v2 9.1 h for 5 folds each (plus a 4 h A/B test of v2 on two folds), including prediction.

3.2 Training setup

Table 11: Training configuration

Hyper-parameter	YOLO26s (experiments)	YOLO26m 5-fold (v1 and v2)
Input size	640 and 960 (upscaled from 352 x 288, letterbox)	960
Epochs / early stopping	100 (640) / 40 (960), patience 30, best-val checkpoint	30 per fold, patience 10, best-val checkpoint of each fold
Batch size	8 (640) / 4 (960), 4 GB VRAM	16 (8 per GPU, 2 GPUs, DDP)
Validation	4 held-out cameras (1066, 1427, 1437, 244)	StratifiedGroupKFold by camera: 5 folds x 3 cameras
Optimizer	Ultralytics "auto" -> AdamW (lr 0.000833, momentum 0.9, wd 0.0005), cosine LR, warm-up 3 epochs	same
Augmentation	Mosaic (off for the last 10 epochs), scale 0.5, translate 0.1, flip 0.5, HSV (h 0.015, s 0.7, v 0.4); applied on the fly, re-sampled every time an image is loaded (no rotation, shear, vertical flip, mixup or copy-paste)	v1: same; v2: same + offline copy-paste and photometric variants (Section 3.3)
Class balance	Repeat-factor sampling (Section 2.6)	same
Seed	42, deterministic=True	same
Hardware	RTX 3050 Ti Laptop GPU (4 GB), 32 GB RAM	Kaggle Notebook, 2 x NVIDIA T4 (16 GB each)

Camera K-fold cross-validation. The single 4-camera split of Section 2.6 wastes 4 cameras for training and gives a noisy score. For the final model the 15 training cameras were split with scikit-learn StratifiedGroupKFold (group = camera, stratified on the rarest class of each image) into 5 folds of 3

cameras. Each fold model is trained on 12 cameras and early-stopped on its own 3 unseen cameras, so every camera is used for validation exactly once (out-of-fold score) and for training in 4 of the 5 models. The folds are fixed in the notebook because a different scikit-learn version produces a different assignment.

3.3 v2: cross-camera copy-paste of rare classes

The CV of v1 showed that rare classes do not transfer between cameras (Songthaew from camera 1426 is not detected on camera 1066). v2 therefore adds training data built **per fold from the training cameras only**:

- **Cross-camera copy-paste:** a crop of a Bus, Tuktuk, Van, Pickup or Songthaew from one camera is pasted into an image of a different camera, in the same lane as (or next to) an existing vehicle, scaled to that vehicle (x the typical size ratio of the two classes), at most 1.6x upscaling, never overlapping a labelled box, with a feathered edge; each source crop is used at most 3 times. 500 synthetic images per fold (~600 pasted vehicles).
- **Photometric variants** (lower contrast, gamma, light blur, JPEG noise) for the repeat-factor copies and the synthetic images, because the test cameras are darker and have lower contrast (Section 2.2).
- 8 non-motorcycle sliver boxes found by the label audit are removed.
- Validation images and labels are untouched; models, folds, epochs and inference are identical to v1.

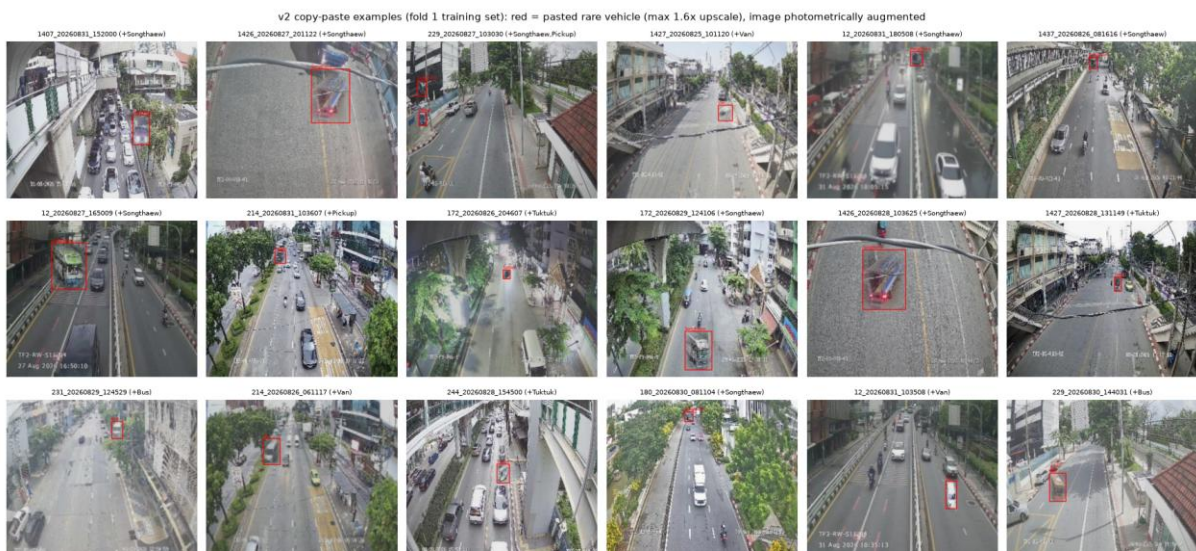


Figure 19: v2 copy-paste examples (red = pasted rare vehicle)

A first A/B test of v2 on folds 1-2 (before the reuse cap was added) was kept only as evidence for the decision; all five v2 fold models used for the submission were trained with the same final code.

3.4 Inference, ensemble and post-processing

- Confidence threshold 0.001: low-confidence boxes only extend the precision-recall curve and never reduce AP, so a very low threshold gives the highest mAP.
- Test-time augmentation (flips + scales) for every model.
- **Weighted Boxes Fusion (WBF)** of the fold models: boxes of the same class from different models that overlap are merged into one box whose coordinates are the confidence-weighted average; the score is averaged over the models, so a box found by only one model gets a low score. The first submission used IoU 0.55 / avg; tuning on the OOF predictions of folds 1-2 showed that 0.55 also

merges neighbouring boxes of the same model, so the final setting is **IoU 0.7, conf_type box_and_model_avg** (Section 4.3). Up to 300 boxes per image are kept.

- **Final ensemble:** 10 models = v1 folds 0-4 + v2 folds 0-4 (experiments/ensemble_wbf.py).
- Boxes are written in absolute pixel coordinates (x1, y1, x2, y2); image_id is the test file name; images without detections get no row and no dummy boxes are added (Data-page rule).

3.5 Evaluation

Both notebooks re-implement the Kaggle metric with pycocotools (COCOeval, IoU = 0.50, area = all, maxDets = 100) on held-out cameras, using the raw labels. Ultralytics' own mAP50 is a bit higher because it uses the cleaned labels.

Chapter 4: Results

4.1 Experiments with YOLO26s (validation = 4 held-out cameras)

Table 12: YOLO26s experiments

Run	Description	Val mAP@50	Kaggle public
yolo26s_640	YOLO26s, imgsz 640, batch 8, RFS; 100 ep max, early-stopped at 60 (best ep 30)	0.4666	-
yolo26s_640_tta	same weights + test-time augmentation (flips + multi-scale)	0.4919	-
yolo26s_960	YOLO26s, imgsz 960, batch 4, RFS; 40 ep (best ep 22)	0.4958	-
yolo26s_960_tta	yolo26s_960 weights + TTA	0.5247	0.5165
yolo26s_960_full_tta	final: same settings trained on all 15 cameras (25 ep, last epoch) + TTA; val cams are inside training, so no unbiased val score	-	0.5481

Higher resolution (640 -> 960) and TTA each add about +0.03. The YOLO26s results below are for the best single-split model (960 + TTA).

Table 13: Per-class AP@50, YOLO26s 960 + TTA, 4 validation cameras

Class	Car	Motorcycle	Bus	Truck	Tuktuk	Van	Pickup	Songthaew
AP@50	0.901	0.800	0.699	0.588	0.448	0.337	0.315	0.109

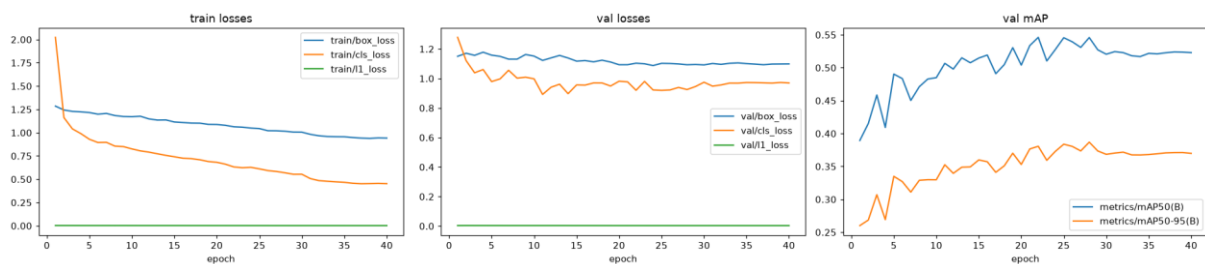


Figure 20: YOLO26s 960: training curves (losses and validation mAP)

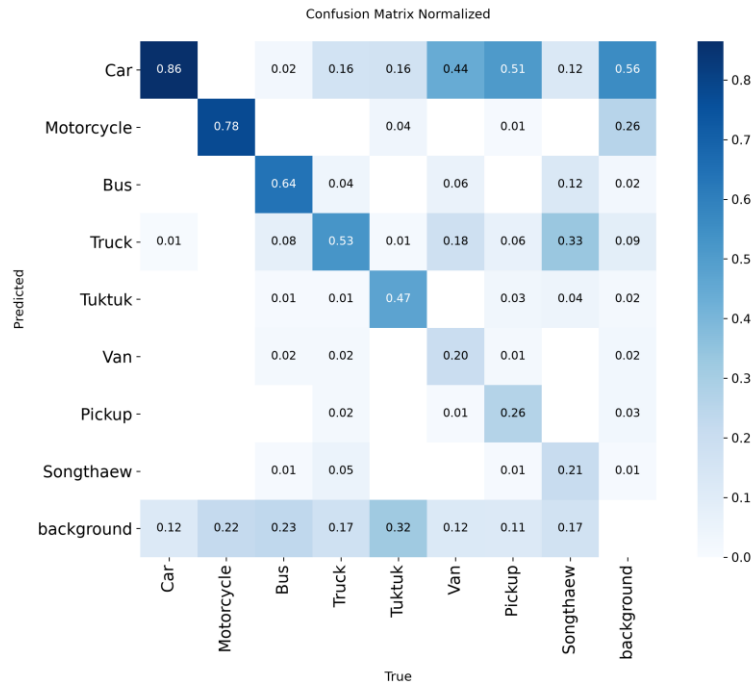


Figure 21: YOLO26s 960: normalised confusion matrix on the validation cameras



Figure 22: YOLO26s 960: ground truth (left) vs prediction with confidence > 0.3 (right)

4.2 YOLO26m v1: camera 5-fold cross-validation

Table 14: YOLO26m 960 + TTA: per-fold and out-of-fold (OOF) validation mAP@50

Fold	Validation cameras	Val mAP@50	Epochs run	Best epoch	Hours
0	1426, 1427, 172	0.4297	28	18	0.95
1	1066, 182, 222	0.5198	30	17	1.84

2	229, 232, 244	0.5025	25	15	1.56
3	1407, 1437, 180	0.7089	30	24	1.85
4	12, 214, 231	0.5449	29	19	1.78
OOF	all 15 cameras (2,991 images)	0.4756			8.0

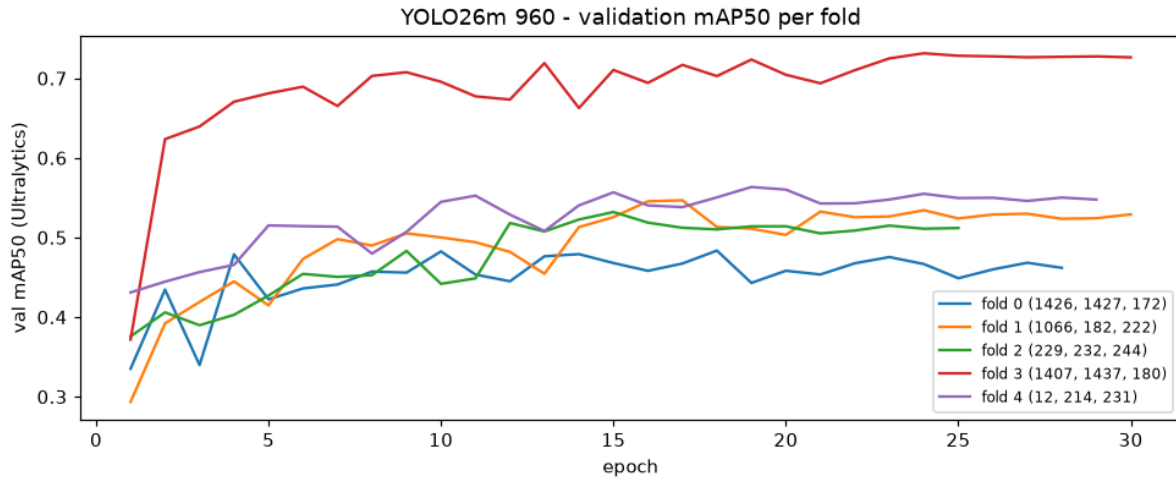


Figure 23: YOLO26m: validation mAP50 per epoch for each fold

Table 15: Per-class AP@50 on cameras that were not used for training

Class	YOLO26s (4 cams)	YOLO26m CV (same 4 cams)	YOLO26m CV (15 cams)
Car	0.901	0.921	0.866
Motorcycle	0.800	0.827	0.694
Bus	0.699	0.672	0.675
Truck	0.588	0.602	0.496
Tuktuk	0.448	0.480	0.483
Van	0.337	0.400	0.300
Pickup	0.315	0.346	0.290
Songthaew	0.109	0.005	0.001
mAP@50	0.525	0.532	0.476

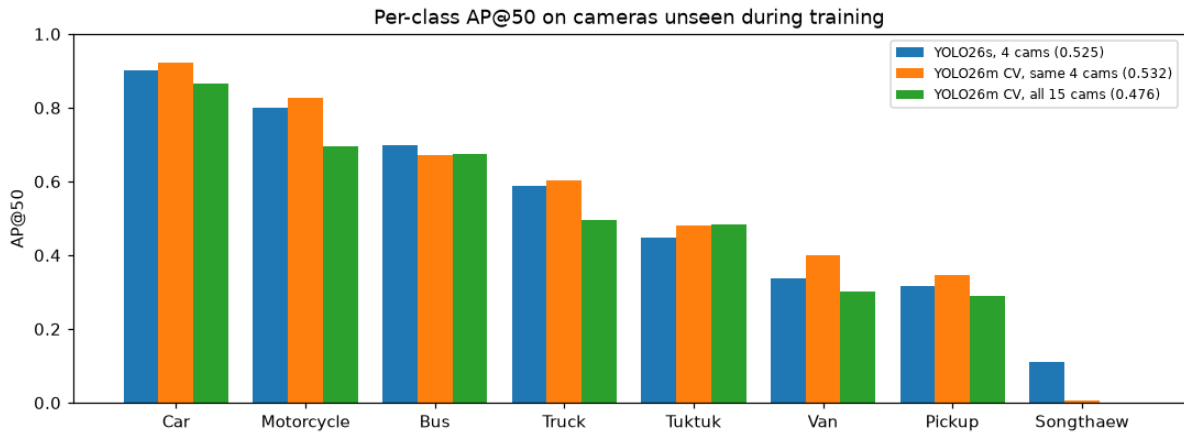


Figure 24: Per-class AP@50: YOLO26s vs YOLO26m on unseen cameras

On the same four cameras YOLO26m is better for 6 of 8 classes (e.g. Van 0.34 -> 0.40, Car 0.90 -> 0.92). The OOF score over all 15 cameras (0.476) is lower mainly because of Songthaew (AP 0.001). 91 of its 117 boxes come from camera 1426. In fold 0 (1426 in validation) only 26 Songthaew boxes are left for training; but even in fold 1, where 1426 is in the training set, Songthaew on camera 1066 is not detected (AP 0.004, 13 boxes). The model ties Songthaew to the close-up view of camera 1426 and does not transfer it to other cameras (see Chapter 5).

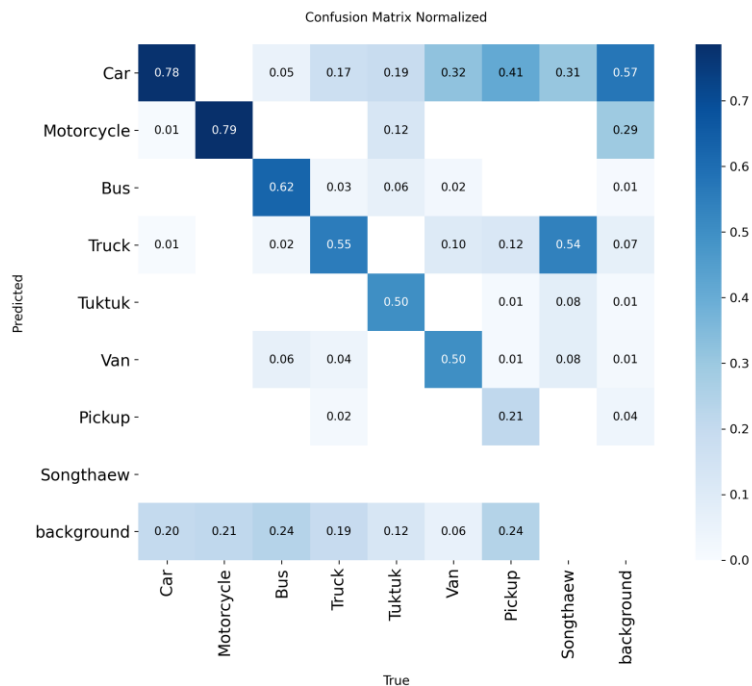


Figure 25: YOLO26m fold 1: normalised confusion matrix on its validation cameras (1066, 182, 222)

4.3 YOLO26m v2 and the v1 + v2 ensemble

Table 16: Validation mAP@50 per fold (each model never saw the validation cameras of its fold)

Fold	v1	v2	v1 (WBF 0.7)	v1 + v2 (WBF 0.7)
0	0.4297	0.4461	0.4297	0.4809
1	0.5198	0.5117	0.5196	0.5527
2	0.5025	0.5018	0.5031	0.5341

3	0.7089	0.6981	0.7095	0.7412
4	0.5449	0.5519	0.5453	0.5777
mean	0.5412	0.5419	0.5414	0.5773

Table 17: Out-of-fold per-class AP@50 over all 15 cameras (WBF IoU 0.7)

Class	v1	v2	v1 + v2
Car	0.866	0.874	0.882
Motorcycle	0.694	0.659	0.694
Bus	0.675	0.600	0.684
Truck	0.496	0.484	0.527
Tuktuk	0.483	0.516	0.572
Van	0.301	0.296	0.337
Pickup	0.291	0.300	0.333
Songthaew	0.001	0.000	0.001
OOF mAP@50	0.4758	0.4662	0.5035

v2 alone is on par with v1 (mean fold mAP 0.542 vs 0.541): copy-paste improves Tuktuk (0.48 -> 0.52) but slightly hurts Bus and Motorcycle, and does not fix Songthaew. Fusing v1 and v2, however, improves **every fold** (+0.03 on average) and the OOF mAP from 0.476 to 0.503: the two runs make different errors, so the ensemble gains on Tuktuk (0.57), Van, Pickup and Truck.

WBF parameter tuning (OOF, folds 1-2)

Table 18: Mean mAP@50 of folds 1-2 for different WBF settings

IoU threshold	conf_type	v1 alone	v1 + v2
0.55	avg	0.4964	0.5245
0.55	box_and_model_avg	0.4964	0.5265
0.55	max	0.5151	0.5230
0.7	avg	0.5113	0.5386
0.7	box_and_model_avg	0.5113	0.5388
0.7	max	0.5111	0.5182
0.8	avg	0.5115	0.5390
0.8	box_and_model_avg	0.5115	0.5391
0.8	max	0.5111	0.5151

IoU 0.55 lowers even a single model (0.511 -> 0.496) because it merges neighbouring vehicles of the same class; IoU 0.7 with box_and_model_avg is the best setting and was used for the final submissions.

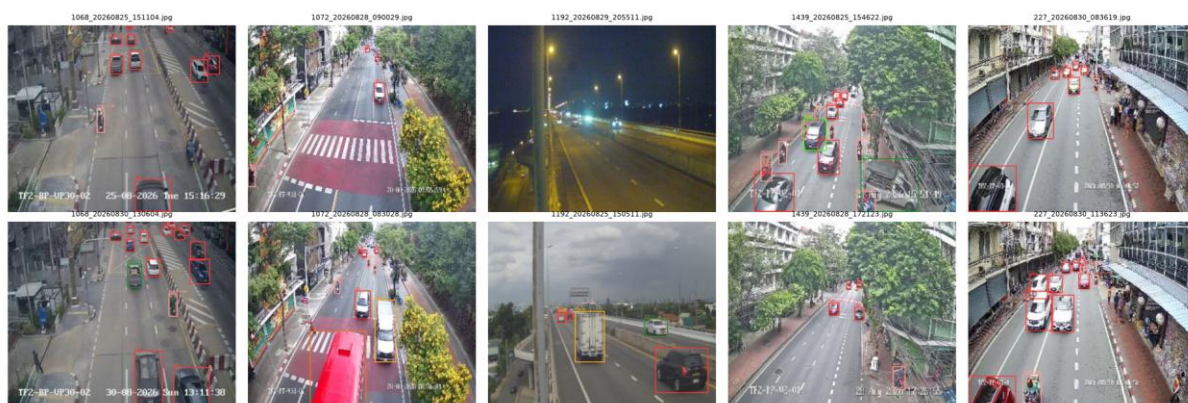


Figure 26: Final v1 + v2 ensemble predictions (confidence > 0.3) on the unseen test cameras

4.4 Kaggle submissions

Table 19: Kaggle submission history

Submitted file	Model / image_id format	Public mAP@50
yolo26s_960_full_tta.csv	YOLO26s 15 cams; long ids from sample_submission	0.00000
yolo26s_960_tta.csv	YOLO26s 11 cams; long ids from sample_submission	0.00000
yolo26s_960_tta.csv (re-submitted)	YOLO26s 11 cams; test file names	0.51648
yolo26s_960_full_tta.csv	YOLO26s 15 cams; test file names, 1,013 images	0.54814
yolo26m_960_cv5fold_wbf.csv	YOLO26m v1, 5 folds, WBF IoU 0.55 / avg	0.60457
yolo26m_960_cv5fold_wbf_iou07.csv	same v1 predictions, WBF IoU 0.7 / box_and_model_avg	0.61367
yolo26m_960_v2_5models_wbf07.csv	YOLO26m v2 (copy-paste), 5 folds, WBF 0.7	0.60237
yolo26m_960_v1v2_10models_wbf07.csv (final)	v1 + v2, 10 fold models, WBF 0.7	0.62963

The first two submissions scored 0.000 although the model was the same: they used the long ids copied from sample_submission.csv, which do not exist in the solution file. Re-submitting the same predictions with the test file names gave 0.516, close to the validation score of the same model (0.525), which confirms that camera-held-out validation is a reliable estimate of the leaderboard. Training YOLO26s on all 15 cameras gave 0.548 and the YOLO26m v1 5-fold WBF ensemble 0.605. Re-fusing the same v1 predictions with the tuned WBF setting gave 0.614 without any training, and the **v1 + v2 ensemble of 10 models gave 0.630**. The public scores follow the OOF scores (v2 alone < v1 < v1 + v2), so the model choice was made on cross-validation, not on the leaderboard.

Final submission file: submissions/yolo26m_960_v1v2_10models_wbf07.csv. Public leaderboard mAP@50: 0.6296.

Reproducibility check. Re-running cv_kfold_kaggle.ipynb locally (RTX 3050 Ti) with the submitted fold weights reproduced every fold score within 0.0002 of the Kaggle run (e.g. fold 1: 0.5198 on Kaggle vs 0.5200 locally) and the ensemble file differs only by floating-point noise (6,948 vs 6,949 boxes with confidence > 0.3).

Model weights of the final ensemble: weights/yolo26m_960_cv_fold{0..4}.pt (v1) and weights/yolo26m_960_cv_v2_fold{0..4}.pt (v2).

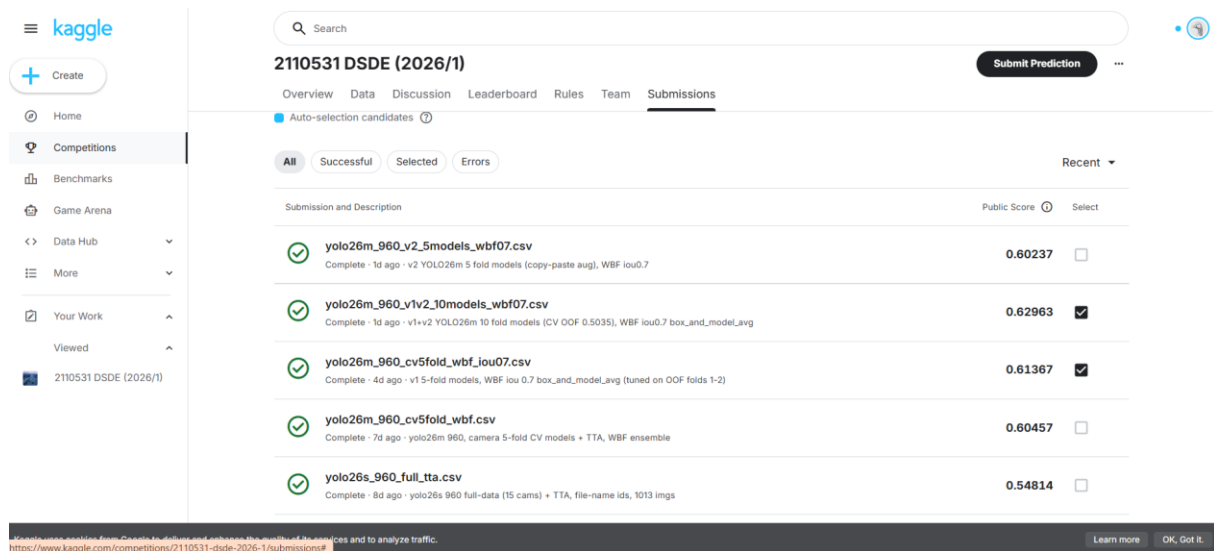


Figure 27: Kaggle submission page (06/10/2026): the two selected final submissions are v1 + v2 (0.62963) and v1 with tuned WBF (0.61367)

Chapter 5: Discussion

5.1 Error analysis by class (YOLO26s, 4 validation cameras)

- **Car (0.90) and Motorcycle (0.80)** are detected well: they are frequent and look distinctive.
- **Van (0.34), Pickup (0.32) and Songthaew (0.11)** are the main weakness. The confusion matrix shows that 44% of Vans and 51% of Pickups are predicted as Car, and 33% of Songthaews as Truck. At 15-25 px these vehicles differ only by a few pixels (roof shape, open cargo bed, rear bench).
- **Label noise** makes this harder: the crops in Chapter 2 contain SUVs labelled as Van and vans labelled as Truck, and 280 vehicles carry two boxes with different classes (mostly Car + Truck). One of the two boxes can never be matched by a single prediction, which caps the achievable AP.
- **Songthaew generalises poorly** because 91 of its 117 training boxes come from one camera (1426), so the model learns that camera's appearance rather than the vehicle.
- **Tuktuk** is often missed completely (32% predicted as background): tuktuks are small, partially covered by motorcycles and look like a motorcycle with a roof.
- The label audit (Section 2.5) confirms the label noise: 674 confident model detections have no label (many are vehicles cut by the image border) and 1,265 labels disagree with a confident model class, about three quarters of them looking like real label errors in a visual check (vans labelled Truck/Bus, pickups labelled Car). These count as false positives / misses for any model and cap the achievable mAP.

5.2 Error analysis by condition and camera

- Day vs night (960 + TTA): 0.537 vs 0.494. Headlight glare and motion blur reduce the visible shape.
- Per validation camera: 0.52 (cam 244, 1427) to 0.69 (cam 1437). Accuracy depends strongly on the viewpoint, so the unseen test cameras are the biggest source of uncertainty.
- Overfitting to the training cameras: validation mAP peaks early (epoch 30 at 640, epoch 22 at 960) and then slowly decreases while the training loss keeps going down. Early stopping on held-out cameras was therefore essential.

- Validation vs leaderboard: the same model scored 0.525 on the held-out cameras and 0.516 on the public test set, so the camera-held-out split predicts the leaderboard well; a random split would have been over-optimistic.

5.3 What helped

- Camera-held-out validation: an honest estimate for unseen cameras (validation 0.525 vs public 0.516 for the same model), and the public scores of all YOLO26m submissions follow their OOF scores.
- Higher input resolution (640 -> 960): +0.029 mAP on validation, mainly Truck and Tuktuk; test-time augmentation: about +0.03.
- Training on all cameras: YOLO26s public 0.516 (11 cams) -> 0.548 (15 cams).
- Larger model + camera 5-fold CV + WBF (YOLO26m v1): public 0.548 -> 0.605.
- Tuning WBF on OOF predictions (IoU 0.55 -> 0.7, box_and_model_avg): public 0.605 -> 0.614 with no training.
- Ensembling two differently augmented CV runs (v1 + v2, 10 models): OOF 0.476 -> 0.503, better in all 5 folds; public 0.614 -> 0.630.
- Very low confidence threshold (0.001) so the precision-recall curve is complete.

5.4 How to improve

- Even larger models (YOLO26l/x, RT-DETR) and imsz 1280, and more ensemble members (several seeds per fold); YOLO26m at 960 already filled the free Kaggle session budget.
- Songthaew: copy-paste augmentation of Songthaew crops from camera 1426 into other cameras, so that the model does not tie the class to one viewpoint.
- Relabel the cases found by the OOF label audit (experiments/audit_*.csv) and retrain; or use the OOF predictions as pseudo-labels for missing border vehicles.
- A second-stage classifier (or a VLM) that re-classifies crops of the confusable classes Car / Van / Pickup / Truck / Songthaew at higher resolution, emitting the alternative classes with lower scores.
- Ensembling several models / resolutions with Weighted Boxes Fusion.
- Semi-supervised domain adaptation: pseudo-label the unlabelled test cameras with high-confidence predictions and fine-tune.
- Copy-paste augmentation of rare vehicles into other cameras, and relabelling the noisy Van / Truck / duplicate boxes.
- Cross-validation over several camera folds to choose hyper-parameters more reliably than a single 4-camera split.

5.5 Lessons from cross-validation

- Songthaew does not generalise across cameras. 91 of its 117 boxes come from camera 1426 (close-up view). In fold 0 (1426 in validation) only 26 Songthaew boxes remain for training (AP 0.009); in fold 1, where 1426 is in the training set, the 13 Songthaews of camera 1066 are still not detected (AP 0.004); folds 3-4 have no Songthaew in validation. The OOF AP of Songthaew is therefore ~0 and the OOF mAP over 15 cameras (0.476) is pessimistic. Copy-paste of Songthaew into other cameras (v2) was tried and did not fix this (Section 5.6).
- Validation mAP varies strongly between folds (0.43 to 0.71): the difficulty depends on the viewpoint, which confirms that a single 4-camera split is noisy and that K-fold CV gives a more reliable model selection.
- Early stopping per fold chose epochs 15-24, so a fixed number of epochs would be sub-optimal for some camera subsets.

- Reproducibility: StratifiedGroupKFold returned a different camera assignment on the Kaggle image than locally (different scikit-learn version); the folds of the submitted run are therefore fixed in the notebook.

5.6 What did not help (negative results)

- Cross-camera copy-paste (v2) alone did not beat v1 (OOF 0.466 vs 0.476; public 0.602 vs 0.614). It helped Tuktuk but slightly hurt Bus/Motorcycle, and Songthaew stayed at AP ~0: pasted rectangular crops do not carry enough context, and there are too few distinct Songthaew vehicles (117 boxes, mostly the same camera).
- Its value was diversity: v2 makes different errors from v1, which is why the v1 + v2 ensemble is the best model.
- Keeping 300 instead of 100 boxes per image (+0.001) and WBF with conf_type max (worse than avg) had no useful effect.

Chapter 6: Conclusion

A YOLO26 detector pretrained on COCO was fine-tuned to detect eight vehicle classes in low-resolution Bangkok CCTV images. The EDA showed that the test cameras never appear in training, that frames of one camera are near-duplicates, that the objects are tiny and that the classes are strongly imbalanced; the pipeline was designed around these facts: camera-held-out validation, repeat-factor sampling for rare classes, upscaled training (imgsz 960) and test-time augmentation.

The final solution trains YOLO26m with camera 5-fold cross-validation on Kaggle GPUs twice (v1 standard augmentation, v2 + cross-camera copy-paste) and fuses the 10 fold models with Weighted Boxes Fusion tuned on out-of-fold predictions. It reaches an out-of-fold mAP@50 of 0.503 on the 15 training cameras and **0.630 mAP@50** on the public leaderboard (from 0.516 for the first single model). The remaining errors are mostly fine-grained confusion between Car, Van, Pickup, Truck and Songthaew, label noise, and viewpoint changes between cameras.

Everything is reproducible from the notebooks with fixed seeds, fixed folds, pinned package versions and the submitted weights; re-running the fold models locally reproduced the Kaggle fold scores within 0.0002.

Appendix: How to reproduce

- `conda create -n dsde python=3.12; pip install torch==2.8.0 torchvision==0.23.0 --index-url https://download.pytorch.org/whl/cu126; pip install -r requirements.txt`
- `kaggle competitions download -c 2110531-dsde-2026-1 -p data/raw`, then unzip into data/raw
- Open vehicle_detection.ipynb from the midterm/ folder and run all cells. The settings are in the Config cell (or set environment variables, e.g. MODEL, IMGSZ, EPOCHS, FULL_TRAIN=1).
- YOLO26s submission without training: `SKIP_TRAIN=1 WEIGHTS=<submitted .pt>` (and the same IMGSZ / TTA as in the submission .json file).
- Final YOLO26m ensemble: run cv_kfold_kaggle.ipynb on a Kaggle Notebook (GPU T4 x2, attach the competition data) or locally. To reproduce without training, copy `weights/yolo26m_960_cv_fold{k}.pt` to `cv_out/fold{k}/best.pt`; the notebook then only predicts and writes `cv_out/fold{k}/test_pred.csv` (PRED_BATCH=8 on a 4 GB GPU). Same for v2 (`cv_kfold_kaggle_v2.ipynb, weights/yolo26m_960_cv_v2_fold{k}.pt -> cv_out_v2/fold{k}/best.pt`).

- Final CSV: python experiments/ensemble_wbf.py
submissions/yolo26m_960_v1v2_10models_wbf07.csv 0.7 box_and_model_avg
experiments/kaggle_cv_run/cv_out experiments/kaggle_cv_run_v2/cv_out_v2